

# Meeting 11 Workstation Activity — Answer Key [50 pts]

## Station 1: Galaxy Rotation Curves [20 pts]

**Warm-up [2 pts]:** SLOWER. If only visible mass exists,  $v(r) = \sqrt{GM/r}$ , so  $v$  decreases as  $r$  increases. **(a) [5 pts]** Area of disk within 8.58 kpc:

$$A = \pi(8.58)^2 = 231.27 \text{ kpc}^2$$

Number of stars:  $N = 2.3 \times 10^8 \times 231.27 = 5.32 \times 10^{10}$ .

$$M_{\text{visible}} = 5.32 \times 10^{10} \times 1.2 M_{\odot} = \boxed{6.38 \times 10^{10} M_{\odot}}$$

**(b) [6 pts]** Convert  $r$  to SI:  $r = 8.58 \times 3.086 \times 10^{19} = 2.648 \times 10^{20} \text{ m}$ .

$$M_{\text{actual}} = \frac{v^2 r}{G} = \frac{(3.4 \times 10^5)^2 (2.648 \times 10^{20})}{6.674 \times 10^{-11}} = 4.59 \times 10^{41} \text{ kg}$$

$$\boxed{M_{\text{actual}} \approx 2.31 \times 10^{11} M_{\odot}}$$

**(c) [7 pts]**

$$M_{\text{dark}} = 2.31 \times 10^{11} - 6.38 \times 10^{10} = \boxed{1.67 \times 10^{11} M_{\odot}}$$

$$\text{Ratio} = \frac{2.31 \times 10^{11}}{6.38 \times 10^{10}} \approx \boxed{3.6 \text{ times heavier}}$$

$$\text{Percent dark matter} = \frac{1.67 \times 10^{11}}{2.31 \times 10^{11}} \approx \boxed{72\%}$$

## Station 2: Gravitational Lensing [15 pts]

**(a) [4 pts]**

$$\boxed{M = \frac{\theta b c^2}{4G}}$$

**(b) [3 pts]**

$$\theta = \frac{0.0068}{4.789} = \boxed{1.42 \times 10^{-3} \text{ rad}}$$

(c) [8 pts] Convert  $b$ :  $b = 0.232 \times 3.086 \times 10^{22} = 7.160 \times 10^{21}$  m.

$$M = \frac{(1.42 \times 10^{-3})(7.160 \times 10^{21})(9.0 \times 10^{16})}{4(6.674 \times 10^{-11})} = \frac{9.15 \times 10^{35}}{2.670 \times 10^{-10}} = 3.43 \times 10^{45} \text{ kg}$$

$$M \approx 1.72 \times 10^{15} M_{\odot}$$

### Station 3: The Fate of Valdris

[15 pts]

(a) [5 pts] Convert  $H_0$  to SI:

$$H_0 = \frac{70.0 \times 10^3 \text{ m/s}}{3.086 \times 10^{22} \text{ m}} = 2.269 \times 10^{-18} \text{ s}^{-1}$$

$$\rho_{\text{crit}} = \frac{3H_0^2}{8\pi G} = \frac{3(2.269 \times 10^{-18})^2}{8\pi(6.674 \times 10^{-11})} = \frac{1.545 \times 10^{-35}}{1.677 \times 10^{-9}}$$

$$\rho_{\text{crit}} \approx 9.21 \times 10^{-27} \text{ kg/m}^3$$

(b) [2 pts]

$$\Omega_m = \frac{\rho_m}{\rho_{\text{crit}}} = \frac{2.76 \times 10^{-27}}{9.21 \times 10^{-27}} = 0.30$$

(c) [5 pts]

$$\Omega_k = 1 - \Omega_m - \Omega_{\Lambda} = 1 - 0.30 - 0.65 = +0.05$$

Since  $\Omega_k > 0$ , Valdris is OPEN. It will expand forever, never recollapsing. (d) [3 pts]  
Ordinary matter is 15% of  $\Omega_m$ :

$$\Omega_{m,\text{ord}} = 0.15 \times 0.30 = 0.045 = 4.5\%$$

This is very close to our own universe's  $\sim 5\%$ . Valdris's ordinary matter fraction is **about the same** as ours.